Database System Proposal

Foodsy

Online Food Delivery Management System

Subject: Database Systems
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Submitted in partial fulfillment of the requirements for the Database Systems course.

Contents

1. Introduction

Foodsy is a proposed online food delivery management system designed to streamline the process of ordering, managing, and delivering food from restaurants to customers. With the increasing demand for digital food services, restaurants require efficient systems to handle customer orders, track deliveries, and maintain structured records.

This system will provide a centralized database that enables customers to place orders online, managers to monitor and assign orders, and employees to fulfill deliveries efficiently. The system aims to improve operational efficiency, reduce manual workload, and support better decision-making through organized and accessible order data.

2. Problem Statement

Many small and medium-sized restaurants currently manage orders manually—through phone calls or informal platforms such as WhatsApp. This approach lacks automation, proper tracking, and data organization, resulting in slow, error-prone order management.

Additionally, restaurant managers lack access to structured data on customer behavior and operational performance. Without centralized records, it becomes difficult to identify popular items, monitor peak order times, or optimize delivery routes. The absence of a proper database system leads to poor decision-making and reduced service quality.

The proposed Foodsy system addresses these challenges by automating and organizing the entire food ordering and delivery workflow through a structured relational database.

3. Objectives

The primary objectives of the Foodsy system are as follows:

1. Automate the food ordering and delivery process.
2. Manage and organize customer orders in a centralized system.
3. Maintain a structured relational database for orders, customers, employees, and menu items.
4. Enable customers to track the real-time status of their orders.
5. Allow managers to assign delivery tasks to employees based on area.
6. Provide employees with a clear view of their assigned deliveries.
7. Generate order-based insights to support managerial decision-making.

4. Scope

The Foodsy system will cover the following functional areas:

1. Admin and manager login with full system management capabilities.
2. Customer registration, login, and profile management.
3. Order placement, cart management, and order tracking.
4. Delivery task assignment to employees based on geographic area.
5. Restaurant menu management (add, update, remove items).
6. Delivery status updates and real-time order tracking.
7. Basic analytics dashboard for managers to monitor performance trends.

The system focuses specifically on the **database layer and operational workflow** of a restaurant's delivery system. Frontend development and payment gateway integration are outside the scope of this project.

5. Data Collection

The data required to operate the system will be sourced from restaurant operations and will include the following entities:

1. **Customer Data:** Full name, contact number, and delivery address.
2. **Employee Data:** Name, employee ID, and assigned delivery area.
3. **Menu Data:** Item names, categories, descriptions, and pricing.
4. **Order Data:** Ordered items, quantities, total amount, order status, and payment method.
5. **Delivery Records:** Delivery timestamps, assigned employee, and completion status.

All data will be stored and managed within a relational MySQL database, providing a structured foundation for system operations and business analysis.

6. Technology Stack

lightgray Component	Technology
Database Management System	MySQL
Query Language	SQL (Structured Query Language)
Database Tool	MySQL Workbench
Development Environment	Local server with SQL interface

7. System Functionality

The system will support the following core functionalities:

1. User registration and login (customers, employees, and managers).
2. Menu browsing for customers.
3. Adding items to a cart and placing food orders.
4. Displaying order summary and itemized bill.
5. Selection of payment method.
6. Order cancellation (permitted before processing begins).
7. Real-time order status tracking by customers.
8. Manager ability to add, update, or remove menu items.
9. Manager ability to assign delivery orders to employees.
10. Employee access to view orders assigned to them.
11. Employee ability to update delivery status upon completion.
12. Basic reporting and order analytics for managers.

8. User Classes

The system will support three distinct user roles, each with specific permissions and responsibilities.

8.1. Manager / Admin

- Manages the overall system and database records.
- Adds, updates, or removes employee accounts.
- Manages and maintains the restaurant menu.
- Assigns delivery orders to available employees.
- Monitors order data, trends, and performance reports.

8.2. Employee

- Logs into the system with assigned credentials.
- Views orders assigned by the manager.
- Delivers orders to the respective customers.
- Updates delivery status upon completion.

8.3. Customer / User

- Registers and logs into the system.
- Browses the restaurant menu and selects items.
- Places orders and receives an order confirmation.
- Tracks the status of their order in real time.

9. Task Division

Sr. No	Name	Responsibilities
1.	Muhammad Bilal	Admin and manager functionalities: system management, menu control, employee assignment, and reporting.
2.	Waseen Ullah	Employee functionalities: delivery management, order assignment, and status updates.
3.	Hassnain Ali	Customer functionalities: registration, order placement, cart management, and order tracking.

10. Conclusion

The Foodsy Online Food Delivery Management System presents a database-driven solution to replace traditional, manual order management in small restaurants. By implementing a structured relational database using MySQL, the system will automate order processing, delivery assignment, and operational recordkeeping.

The system will also equip managers with the data insights needed to analyze order trends and make informed business decisions. Foodsy is designed to be efficient, organized, and scalable—providing a solid foundation for modern food delivery operations.

11. References

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